

Simulation-based cutoffs for Rasch item fit

Iterations, multiplicity correction, and decision stability

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The starting point

Parametric bootstrap gives cutoffs for conditional infit (Müller 2020) that fit the sample and items at hand (Johansson 2025).

- Simulate B datasets that fit the Rasch model, compute infit in each
- Flag an item when its observed infit falls outside an interval of width w
- The earlier guidance was 100 to 400 iterations, scaled by sample size

That guidance was based on single-item detection and error rates. The error rate across the whole scale was never checked.

A better rule from the same bootstrap

Bootstrap p-values with a Westfall–Young step-down ([Westfall and Young 1993](#); [Ferreira 2024](#)) control the familywise error rate (FWER) directly, whatever the number of items.

Iterations	100	400	2000
Uncorrected	62.2%	60.7%	59.8%
Westfall–Young	6.8%	4.3%	3.8%
Benjamini–Hochberg	0.0%	4.2%	4.6%

- Against an interval matched to the same error rate, Westfall–Young FWER detected more misfit in every condition, with slightly fewer false positives
- Benjamini–Hochberg ([Benjamini and Hochberg 1995](#)) cannot flag a single item until $B \geq m/\alpha$, which is 400 for 20 items

How the Westfall–Young step-down works

It asks how unusual the worst item is, compared with the worst item in a scale that fits.

1. **Common scale.** Express each item's infit as distance from its own simulated mean, in simulated SDs
2. **Worst by chance.** In each simulated dataset, record the most extreme item. This shows how extreme the worst item gets when every item fits
3. **Test the worst observed item.** Its p-value is the share of simulated datasets whose most extreme item is at least as extreme
4. **Step down.** Set that item aside and repeat on the rest. Stop at the first item that is not significant

Every item is judged against the worst of many, so the chance of any false flag stays at 5%. The simulated datasets keep the correlations between items, which makes the test less strict than Bonferroni.

Would a second analyst agree?

The same data were analyzed with 25 different seeds (randomness starting points), so any disagreement is simulation noise alone.

Realistic misfit, Westfall–Young FWER rule:

Iterations	100	400	1000	2000
Two analysts disagree on at least one item	22%	10%	6%	4%

- Error rates are stable from 400 iterations, but **reproducibility** keeps improving after that
- Items that stay seed-dependent at high B sit near the decision boundary. Treat them as borderline, not as settled by a luckier seed

Recommendations

1. Flag items on the Westfall–Young FWER corrected p-value at $\alpha = .05$
2. Use at least 400 iterations, and 1000 to 2000 for a final analysis
3. Visualize the interval as a descriptive band at $w = .95$
4. Report the number of iterations and the seed
5. Decide on infit. Outfit detected less in every condition tested

All of this is available in easyRasch2 and the jamovi module easyRasch2jmv.

References

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